

Color Threshold Adjustment

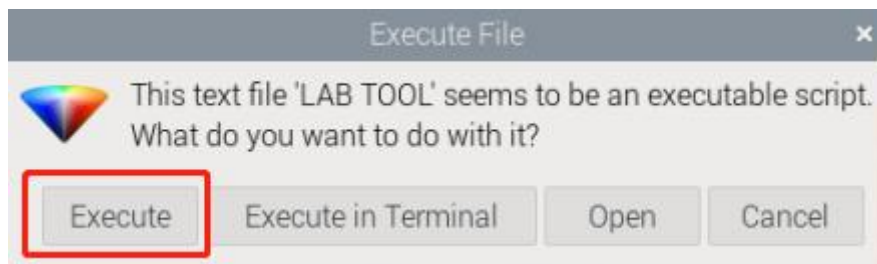
The color of an object can change with the light sources, which can affect the functionality involved color recognition. To tackle this issue, this lesson will introduce you how to use LAB Tool to adjust the color threshold.

1. Enable LAB_Tool

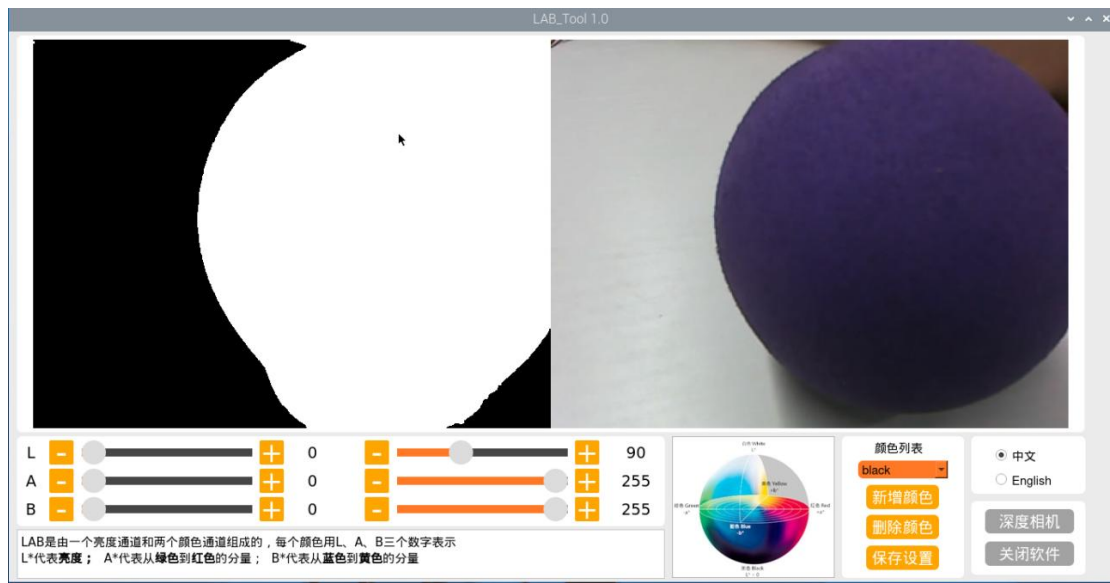
1) Open the VNC software. Locate and open the "LAB_Tool" on the main interface.



2) The "Execute File" dialog box will appear. Click "Execute" to proceed.



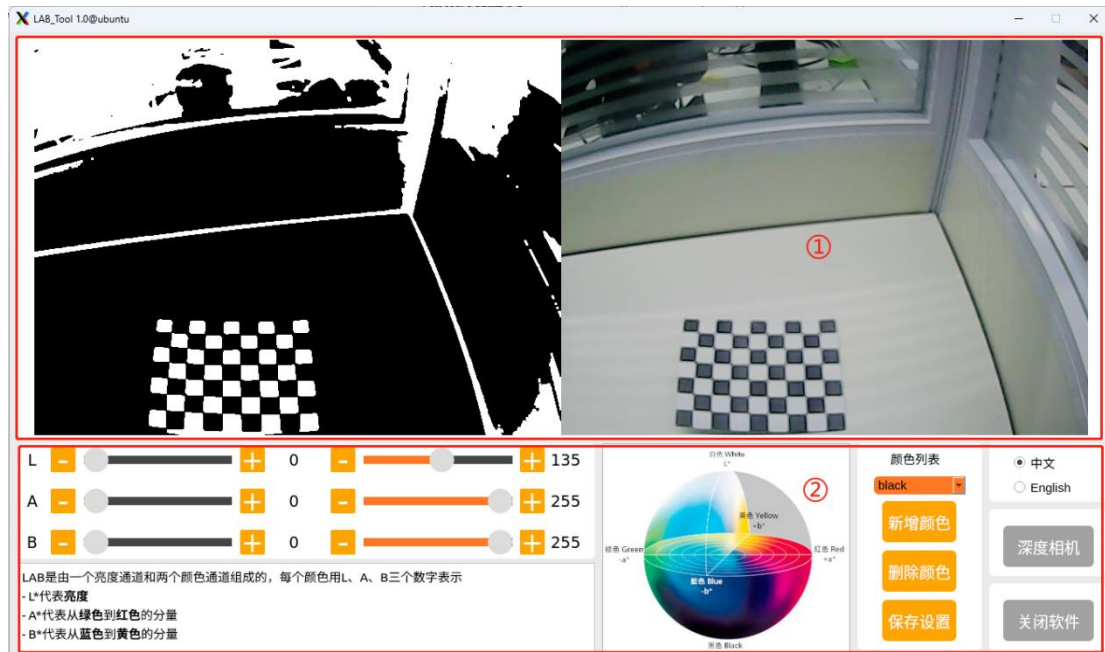
3) After entering the software, you can see the interface of LAB_Tool, as shown in the figure below:



Note: If the live feed camera does not appear in the pop-up interface, it means that the camera is not connected successfully. Please check if the camera connection cable is

properly connected.

The interface of LAB tool software is divided into two parts: image display area and recognition adjustment area.

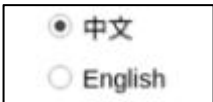


4) Image display area: the left displays the processed image and the right is the raw image.

Note: if the transmitted image does not display normally, it's the issue of camera connection. In this case, you need to examine if the wiring is connected properly or reconnect it.

5) Recognition adjustment area: adjust the color threshold. The function of each button refer to the following table:

Icon	Function instruction
	The sliders L, A and B are used respectively to adjust the values of the corresponding L, A and B components of the image. The sliders on the left are the “min” value for each component while the slider on the right are the “max” value for each component.

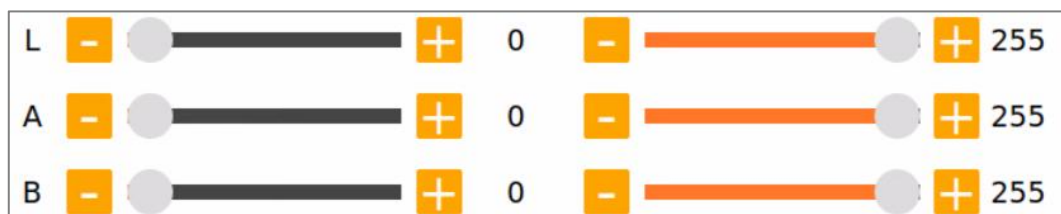
	Choose color to adjust the threshold.
	Delete the currently selected color.
	Add recognizable color.
	Save the adjustment result of the color threshold.
	Click this button to switch between Chinese and English.
	Click on this button to swap Depth camera/monocular camera.
	Close LAB TOOL.

2. Adjust Color Threshold

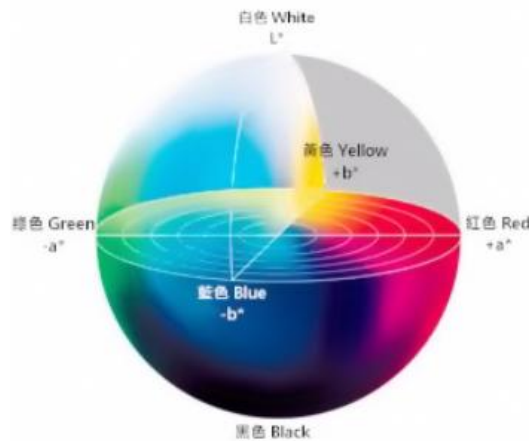
1) Open LAN tool. Choose the color in the color drop-down list of color recognition area. Take a color “red” as example.



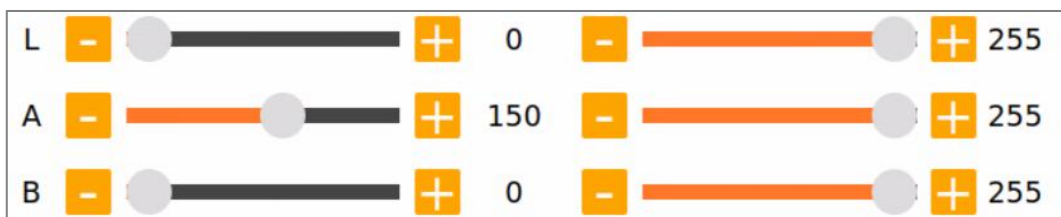
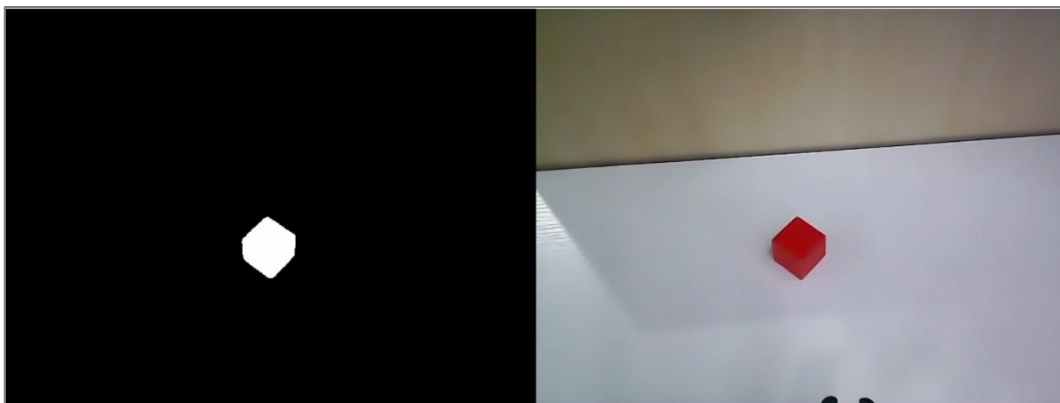
2) Adjust the “min” values of L, A and B components to “0”, and the “max” values to “225”.



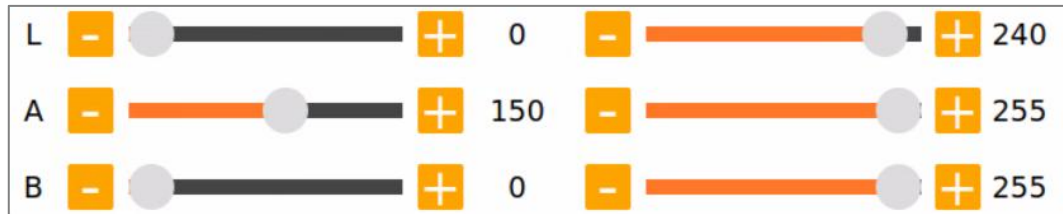
3) Place the target within the camera’s field of view. Adjust the values of L, A and B component towards the interval representing the target recognition color according to the LAB color space distribution chart.



Red color is close to “+a”, which indicates that A component needs to be increased. Therefore, keep the “max” value of the A component unchanged and increase its “min” value until the object on the left turns white while other areas turn black.



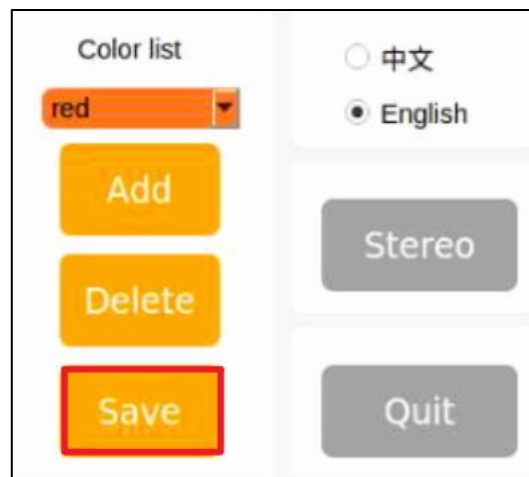
4) Adjust the L and B components according to your surroundings. If the red color is light, increase the “min” value of the L component; if it is dark, decrease the “max” value of the component. If the red color tends to be warm, increase the “min” value of the B component; if it tends to be cool, decrease the “max” value of the B component.



The following table shows the parameter information of LAB threshold adjustment:

Color component	Value Range	Corresponding Color Intervals
L	0~255	Black-white (-L~+L)
A	0~255	green-red (-a ~ +a)
B	0~255	Blue-yellow (-b ~ +b)

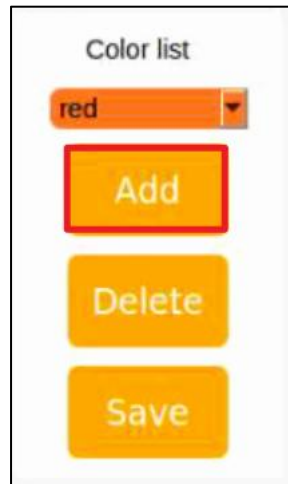
5) Click “Save” button in adjustment area to save the adjustment parameters.



3. Add New Color

In addition to the built-in colors, users can add other recognition colors. Take an example of adding “yellow”.

1) Open LAB tool and click on “Add” button.



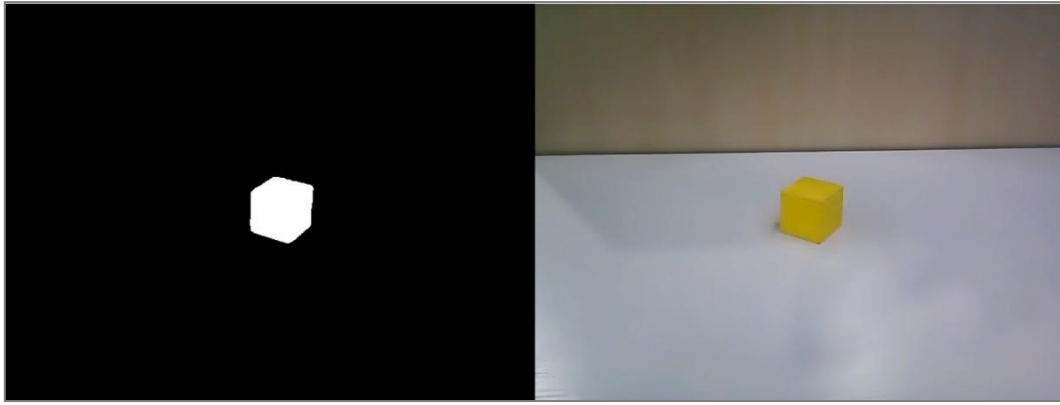
- 2) Fill in the "name" column with the color name, and click the "OK" button.



- 3) Choose the added color in the color drop-down menu.



- 4) Place the target within the camera's field of view, and adjust the threshold by dragging the L, A, and B component sliders until the area of the colored object within the left screen turns white, while the other areas turn black.



L	-		+	0	-		+	200
A	-		+	0	-		+	255
B	-		+	155	-		+	255

Note: the adjustment method of color threshold refers to “2. Adjust Color Threshold”.

5) Click “Save” button in recognition adjustment area to save the adjustment parameters of color threshold.

Color list

yellow

Add

Delete

Save

☐ 中文

☒ English

Stereo

Quit